

Note:

1. You will be penalized for directly copying $\text{\LaTeX}$ code from ChatGPT or any AI tools.
 2. Any case of copying will be treated strictly as per the institute rules.
 3. Submit your solutions in a single zip file **enrollment_number_lab3**.
 4. Match the colors of cross references and links as close as possible to the given ones.
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1. **[8 marks]** Draw the following diagrams using TikZ. Put your TikZ code within the figure environment.

In Figures 1 and 2, we show how to add images in $\text{\LaTeX}$ with TikZ. The figures are taken from [EME22].

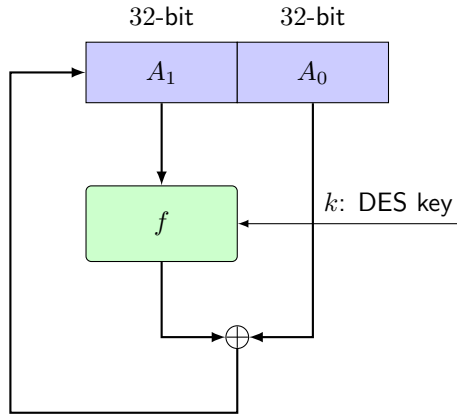


Figure 1: A diagram of word-based feedback shift register

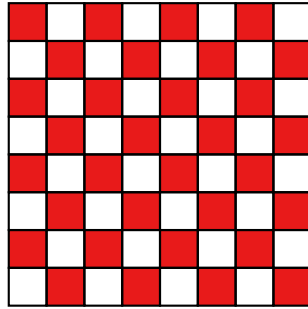


Figure 2: A diagram of checkerboard

2. **[4 marks]** Align the figures as given below.

In Figure 3, we show how to add two figures side by side.

In Figure 4, we show how to use the subfigure environment. The subfigures are shown in Figures 4a and 4b and taken from [HDRM23, Section 1].

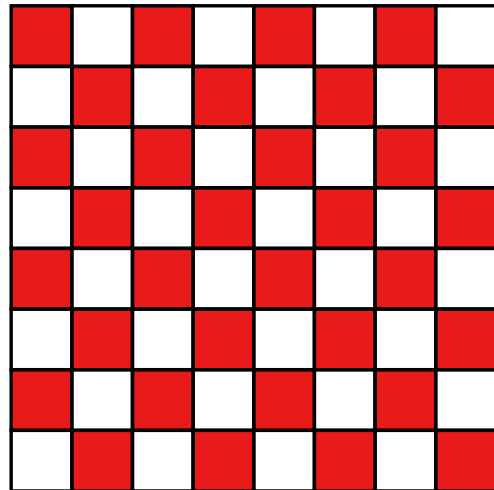
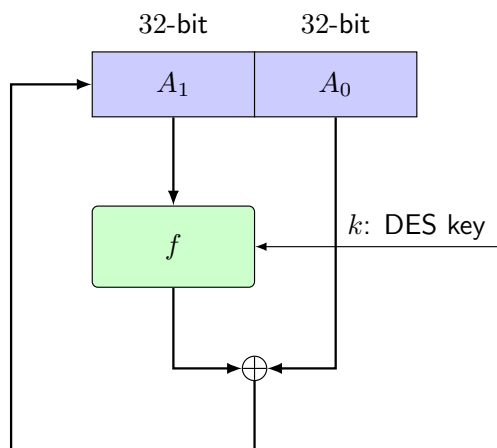
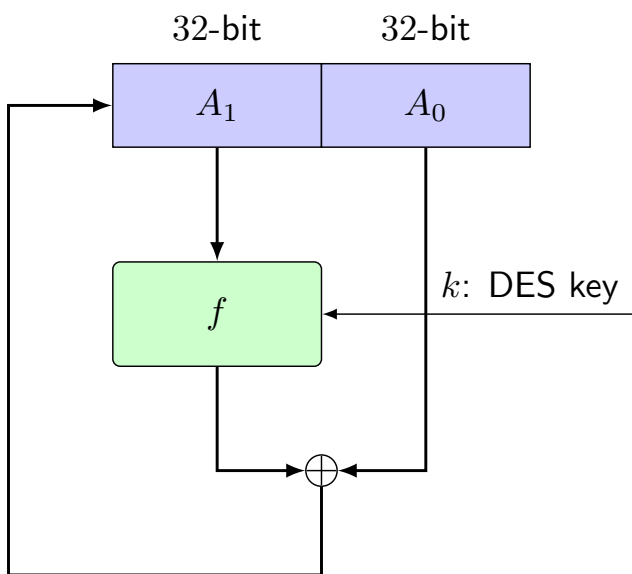
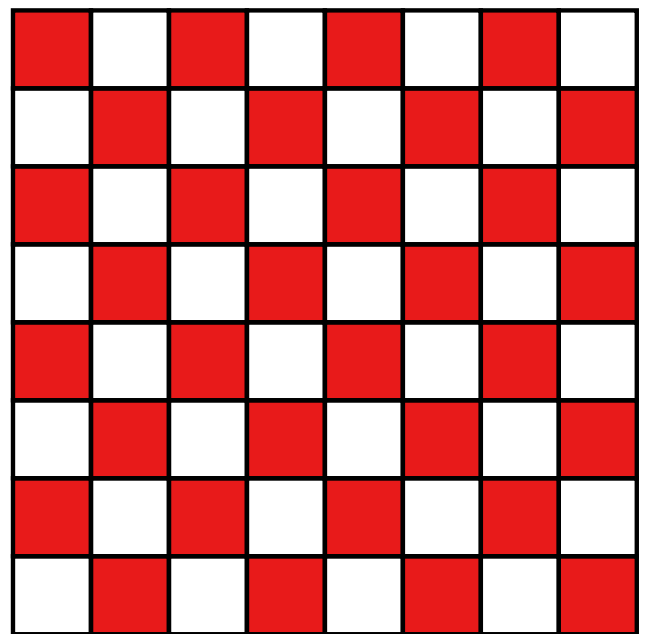


Figure 3: Putting figures side by side



(a) Shift register



(b) Checkerboard

Figure 4: Subfigure environment

3. **[3 marks]** Write the $\text{\LaTeX}$ code for the following.

In the following we show how to use the `minipage` environment and add images in the same environment as shown in Figures 5 and 6. The resources are taken from [Ind26].

Groups

The ubiquitous student groups are what set IIT Roorkee apart from the other technical institutes in India. Here do the like-minded gather to work together toward a certain goal- be it getting out a magazine on time, or putting together a one-hour concert, or making a trip to a nearby village for social work. From togetherness comes success and this feature of life here, certainly adds spice to the lives of the students.

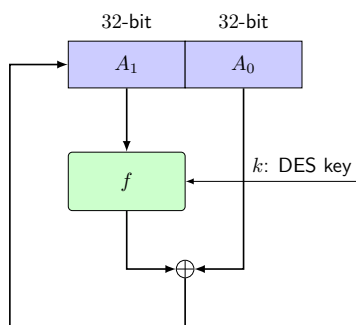


Figure 5: Shift register within Minipage

Fitness

Fit India Mission encourages people to become part of Fit India Movement by inculcating at least 30-60 minutes of physical activities in their day to day lives. The mission of the Movement is to bring about behavioral changes and move towards a more physically active lifestyle.

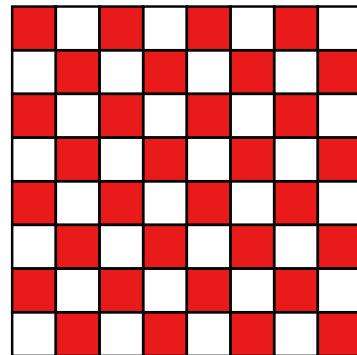


Figure 6: Checkerboard within Minipage

4. **[5 marks]** Create a bib file and match the references exactly as given below.

The references [HDRM23, EME22, Ind26] are used in this lab.

References

- [EME22] Johannes Erlacher, Florian Mendel, and Maria Eichlseder. Bounds for the Security of Ascon against Differential and Linear Cryptanalysis. *IACR Trans. Symmetric Cryptol.*, 2022(1):64–87, 2022. <https://doi.org/10.46586/tosc.v2022.i1.64-87>.
- [HDRM23] Solane El Hirsch, Joan Daemen, Raghvendra Rohit, and Rusydi H. Makarim. Twin Column Parity Mixers and Gaston - A New Mixing Layer and Permutation. In Helena Handschuh and Anna Lysyanskaya, editors, *Advances in Cryptology - CRYPTO 2023 - 43rd Annual International Cryptology Conference, CRYPTO 2023, Santa Barbara, CA, USA, August 20-24, 2023, Proceedings, Part III*, volume 14083 of *Lecture Notes in Computer Science*, pages 475–506. Springer, 2023. https://doi.org/10.1007/978-3-031-38548-3_16.
- [Ind26] Indian Institute of Technology Roorkee. 2026. <https://www.iitr.ac.in/>.